COMPUTER, MATH, AND SCIENCES DEPARTMENT (CMS)

Faculty Members

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Programs Offered

The CMS Department offers two programs – Computer Science and Healthcare Information Systems. Each program grants a Bachelor of Science degree. The details of each program will follow.

Vision

The vision of the Computer, Math, and Sciences (CMS) Department is to achieve academic excellence beyond accreditation standards in computing technologies and Information Systems.

COMPUTER SCIENCE PROGRAM

Mission

The mission of the Computer Science program is to provide students with a state-of-the-art curriculum; maintain enduring relationships with industry to create opportunities for students; cultivate learners with a sense of responsibility towards the profession and a passion for lifelong learning; and provide educational experiences that enable our students to be innovative leaders.

Objectives

The educational objectives of the Computer Science program intend to enable its graduates, within a few years of graduation, to:

- Solve computing problems of varying complexity that meet local, regional, or global markets
- Demonstrate professional skills and ethical behavior in practicing their profession
- Progress in computing and other professionally related fields at the level of professional employment or graduate studies

Program Outcomes

Upon successful completion of the BS program in Computer Science, an RHU graduate in Computer Science should demonstrate the following:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of Computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Accreditation

The Bachelor of Science program in Computer Science is accredited by the Computing Accreditation Commission (CAC) of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Computer Science.

Career Opportunities

A Computer Science major at Rafik Hariri University opens up diverse opportunities and prepares students for careers across various sectors. Our graduates are in industries, governmental and academic units, TV stations, the banking sector, and more. They hold roles such as games developer, system analyst, software engineer, web designer, mobile application developer, database analyst/developer/administrator, network engineer/administrator, data analyst, AI developer, and IT consultant. The work is challenging but intellectually stimulating as you get to work on interesting problems, and it is financially rewarding.

Program Overview

The Computer Science Program at Rafik Hariri University develops students' computational and critical-thinking skills and teaches them how to create, not simply use, new technologies. Students learn to think logically and solve problems by acquiring the fundamental principles of science and advanced techniques currently used in practical systems development. They also learn oral communication and presentational proficiencies that enable them to work with multicultural teams. Furthermore, they learn to work on individual and team projects to develop new systems, thereby enhancing time management and team coordination skills. RHU's Computer Science Department graduates are lifetime learners; they can adapt quickly to this ever-changing, challenging field.

To obtain a Bachelor of Science degree in CS, the student must complete 93 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BS in Computer Science (93 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	6	–	15	9	30	32
College Requirement	–	–	–	–	-	0
Program Requirement	36	9	18	–	63	68
Credits	42	9	33	9	93	100

I. University Requirements (General Education)

The list of the University's required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

III. Program Requirements

A. Mandatory Requirements

Mathematics requirements
The mathematics requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 215	Calculus III	3	
MATH 311	Linear Algebra	3	
MATH 421	Numerical Analysis	3	MATH 311

Business requirements
The business requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
BMGT 300	Project Management	3	Junior Standing

Major requirements
The CS mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 341	Software Engineering	3	COSC 214
COSC 351	Logic Design	3	
COSC 351L	Logic Design Lab	1	Co-req: COSC 351
COSC 353	Computer Organization	3	COSC 351
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req: COSC 360
COSC 399	Co-op Training Experience	1	ENGL 217; Senior Standing
COSC 421	Theory of Computation	3	COSC 215 and MATH 210

COSC 451	Operating Systems	3	COSC 214
COSC 490	Summative Learning Experience	3	ENGL 217

B. Technical Electives

As part of the Bachelor of Science in Computer Science program, students must take 9 credit hours of technical electives. These courses allow students to focus on a specific area for in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various Computer Science disciplines. In cooperation with the academic advisor, the student should select electives that best align with their needs and aspirations. The listed technical elective courses and other courses from engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

1. Internet and Computer Networks
2. Game Design and Development
3. Artificial Intelligence and Data Mining
4. Mobile/Web Design and Development

The CS elective courses are listed in the table below:

Course #	Title	Credits	Prerequisites
BITM 320	Enterprise Resource Planning	3	COSC 231
CCEE 426	Design of Embedded Systems	3	COSC 351
CCEE 561	Computer Vision	3	COSC 214
CCEE 566	Natural Language Processing	3	COSC 214
COSC 434	Advanced Web Programming	3	COSC 231 and COSC 333
COSC 435	Mobile Application Development	3	COSC 214
COSC 436	Game Programming	3	COSC 214
COSC 438	Introduction to Software Testing	3	COSC 341
COSC 461	Advanced Networking	3	COSC 360
COSC 463	Cybersecurity Operations	3	COSC 360
COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
COSC 481	Artificial Intelligence	3	COSC 214
COSC 482	Data Science and Web Scraping	3	Co-req.: COSC 333 and MATH 351
COSC 496	Independent Study I	3	Senior Standing and a CGPA of 80 and above.
COSC 497	Independent Study II	3	Senior Standing and a CGPA of 80 and above.
COSC 498	Special Topics in Computer Science	3	Senior standing
HCIS 341	Health Information Systems I	3	
HCIS 342	Health Information Systems II	3	HCIS 341

Or any other approved CS elective course.

In addition, the students can take the following three labs instead of one elective.

Course #	Title	Credits	Prerequisites
Networking			
COSC 460L	Networking Lab 2	1	Co-req.: COSC 360L
COSC 461L	Networking Lab 3	1	COSC 460L
COSC 462L	Networking Lab 4	1	Co-req.: COSC 461L or CCEE 555L

Or any other approved CS elective course.

C. Summative Learning Project

Students must complete a 3-credit-hour course (COSC 490 Summative Learning Experience) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or on an innovative idea. The SLP could be an extension of the projects the students had worked on in computing courses, leading to more useful outcomes.

D. Co-op Training Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students must submit a formal report and a poster and make a formal presentation about their co-op experience. While ENGL 217 and senior standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Course Coding

The courses offered in the CS program are designated code numbers in the form of (COSC ABC), where:

a	Designates year/level
b	Designates are as follows: 1. Computer Software; 2. Computer Science Theory; 3. Mobile/Web Development; 4: Systems Analysis and Design; 5: Computer Hardware; 6: Networking and Security
c	Designates the course sequence in an area

Study Plan

The Bachelor of Science in Computer Science encompasses 93 credit hours over six semesters and two summer sessions. The following study plan serves as a roadmap for a smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Semester 1 (15 Credits)			
COSC 214	Introduction to Programming	3	
COSC 351	Logic Design	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
	Natural Sciences and Technology Elective	3	
Year 1, Semester 2 (16 Credits)			
ARAB 212	Arabic Language and Communication	2	
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 351L	Logic Design Lab	1	COSC 351
	Community & Sustainability	1	Co-req,: ENGL 210
ENGL 217	Professional English Communication	3	ENGL 210
MATH 210	Discrete Mathematics	3	
Summer Session (3 Credits)			
	Humanities/Fine Arts Elective	3	
Year 2, Semester 1 (16 Credits)			
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req,: COSC 360
MATH 351	Probability and Statistics	3	MATH 215
	Social Science/Culture Elective I	3	
Year 2, Semester 2 (15 Credits)			
BMKA 200	Introduction to Marketing	3	ENGL 210
COSC 341	Software Engineering	3	COSC 214
MATH 311	Linear Algebra	3	
	CS Major Elective I	3	
	CS Major Elective II	3	
Summer Session (1 Credit)			
COSC 399	Co-op Training Experience	1	ENGL 217; Senior Standing
Year 3, Semester 1 (15 Credits)			
COSC 353	Computer Organization	3	COSC 351

COSC 421	Theory of Computation	3	COSC 215 and MATH 210
COSC 451	Operating Systems	3	COSC 214
GRDS 440	Design and Social Impact	3	Junior Standing
	Social Sciences/Culture Elective II	3	
Year 3, Semester 2 (12 Credits)			
BMGT 300	Project Management	3	Junior Standing
COSC 490	Summative Learning Experience	3	ENGL 217
MATH 421	Numerical Analysis	3	MATH 311
	CS Major Elective III	3	

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 215	**Advanced Programming and Data Structures**	**3(2,2)**
Continuing COSC 214, this course covers the basics of Object-Oriented Programming (OOP) languages and data structures. It covers pointers, classes, encapsulation, constructors/destructors, object instantiation, and templates. The course also details data structures and applications such as stacks, queues, lists, and trees. **Prerequisite***:* COSC 214. Equivalent to CCEE 216.		

COSC 231	**Database Management Systems**	**3(3,0)**
This course exposes students to the fundamental concepts for designing, using, and implementing database systems. It covers database concepts, data modeling, relations, normalization, and data manipulation languages. **Prerequisite***:* COSC 214. Equivalent to CCEE 315.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	Web Programming	3(3,0)
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.		

COSC 341	Software Engineering	3(3,0)
This course provides a deep and comprehensive understanding of the fundamentals of software engineering. The course introduces students to software process models, project management, software requirements, and engineering processes. The course covers SDLC stages, UML diagrams, and the use of CASE tools. **Prerequisite***:* COSC 214. Equivalent to CCEE 310.		

COSC 351	Logic Design	3(3,0)
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices**. Prerequisite***:* None. Equivalent to CCEE 221.		

COSC 351L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits, including combinational circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: COSC 351. Equivalent to CCEE 221L.		

COSC 353	Computer Organization	3(3,0)
This course exposes students to computer system organization and design; implementation of CPU data paths and control; instruction set design; memory hierarchy (caches, main memory, virtual memory) organization and management; input/output subsystems (bus structures, interrupts, DMA); performance evaluation; and pipelined processors. **Prerequisite**: COSC 351. Equivalent to CCEE 324.		

COSC 360	Networking	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate various trade-offs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field of networks. **Prerequisite***:* COSC 214. Equivalent to CCEE 354.		